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Modeling and Discrete Simulation

Subject 1
Introduction

Mujdat Soyturk, Ph.D.
Associate Professor

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Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010" and from textbook's course slides.

Definition

- A *simulation* is the imitation of the operation of real-world process or system over time.
  - Generation of artificial history and observation of that artificial history
- A *model* construct a conceptual framework that describes a system
- The behavior of a system that evolves over time is studied by developing a *simulation model*.
- The model takes a set of expressed assumptions:
  - Mathematical, logical
  - Symbolic relationship between the *entities*

Goal of modeling and simulation

- A model can be used to investigate a wide variety of "what if" questions about real-world system.
  - Potential changes to the system can be simulated, in order to predict their impact on the system performance.
  - Find adequate parameters before implementation
- So simulation can be used as
  - An analysis tool for predicting the effect of changes to existing systems
  - A design tool to predict the performance of new system
- It is better to do simulation before Implementation.

How a model can be developed?

- Mathematical Methods
  - The use of probability theory, algebraic methods,…
  - Their results are accurate
  - They have a few number of parameters (measures)
  - It is **impossible** for complex systems → e.g. real-world systems.
- Numerical computer-based simulation
  - It is simple
  - It is useful for complex system

When Simulation Is the Appropriate Tool (1)

- Simulation enable the study of internal interaction of a subsystem with complex system
- Informational, organizational and environmental changes can be simulated and find their effects
- A simulation model help us to gain knowledge about improvement of the system
- Changing simulation inputs and observing the results can produce valuable insights about variables e.g. importance, relationship, interaction.
- Finding important input parameters with simulation can be used with new design and policies before implementation

When Simulation Is the Appropriate Tool (2)

- Simulations can be used to verify analytical solutions.
- Simulating different capabilities for a machine can help determine its requirements.
- Simulation models designed for training make learning possible without the cost and disruption of the on-the-job instruction.
- A plan can be visualized with animated simulation
- A modern system (factory, wafer fabrication plant, service organization) is so complex that its internal interaction can be treated only through simulation.

When Simulation Is Not Appropriate

- When the problem can be solved by common sense.
- When the problem can be solved analytically.
- If it is easier to perform direct experiments.
- If cost exceed savings.
- If resource or time are not available.
- If there is no data available nor the estimates.
- If the validation and verification are not possible, due to time or personnel unavailability.
- If system behavior is too complex.
  - Like human behavior

Advantages and disadvantages of simulation

- In contrast to optimization models, simulation models are “run” rather than solved.
  - Given as a set of inputs and model characteristics the model is run and the simulated behavior is observed
    - This process of changing inputs and model characteristics results in a set of scenarios that are evaluated.

Advantages of simulation

- New policies, operating procedures, information flows and so on can be explored without disrupting ongoing operation of the real system.
- New hardware designs, physical layouts, transportation systems and so on can be tested without committing resources for their acquisition.
- Hypotheses about how or why certain phenomena occur can be tested for feasibility.
- Time can be compressed or expanded to allow for a speed-up or slow-down of the phenomenon under investigation.
- Insight can be obtained about interaction of variables and important variables to the performance of the system.
- Bottleneck analysis can be performed to discover where work in process, the system is delayed.
- A simulation study can help in understanding how the system operates.
- “What if” questions can be answered.

Disadvantages of simulation

- Model building requires special training.
  - The model developed by different individuals may be different in addition to some similarities.
    - Vendors of simulation software have been actively developing packages that contain models that only need input (templates).
- Simulation results can be difficult to interpret.
  - Results are due to the system itself or of randomness?
    - Many simulation software have output-analysis
- Simulation modeling and analysis can be time consuming and expensive.
    - New models are developed. Advances in simulation packages.
    - Evaluation cost reduction due to cheaper hardware solutions.

Areas of application (1)

- Please check:
  - The Winter Simulation Conference, WSC.
  - Micro Saint Sharp website; whitepapers.
  - Anylogic website; cases.
- Manufacturing applications
- Semiconductor manufacturing
- Construction engineering and project management
- Military application
- Logistics, supply chain and distribution application
- Transportation modes and traffic
- Business Process Simulation

Areas of application (2)

- Health Care
- Automated Material Handling System (AMHS)
  - Test beds for functional testing of control-system software
- Risk analysis
  - Insurance, portfolio,...
- Computer Simulation
  - CPU, Memory,…
- Network simulation
  - Internet backbone, LAN (Switch/Router), Wireless, PSTN (call center),...

Systems and System Environment

- A *system* is defined as a groups of objects that are joined together in some regular interaction toward the accomplishment of some purpose.
  - An automobile factory: Machines, components parts and workers operate jointly along assembly line
- A system is often affected by changes occurring outside the system: system environment.
  - Factory : Arrival of orders
    - Effect of supply on demand : relationship between factory output and arrival (activity of system)
  - Banks : arrival of customers

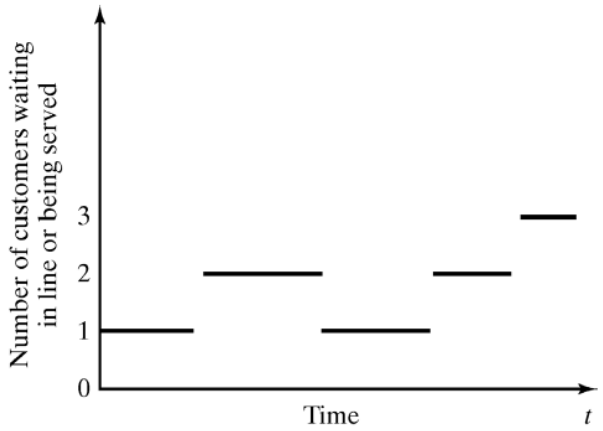
*Decide on the boundary between the system and the system environment

Components of system

- Entity
  - An object of interest in the system : in factory: machines - in bank: customers.
- Attribute
  - The property of an entity : in factory: speed, capacity - in bank: the balance in their accounts
- Activity
  - A time period of specified length: in factory: welding,stamping - in bank: the making deposits
- State
  - A collection of variables that describe the system in any time : in factory: status of machine (busy, idle, down,…) - in bank: the number of busy tellers, customers waiting in the line
- Event
  - A instantaneous occurrence that might change the state of the system: in factory: breakdown - in bank: arrival of a customer
  - Endogenous
    - Activities and events occurring with the system: in bank: completion of service of a customer
  - Exogenous
    - Activities and events occurring with the environment: in bank: arrival of a customer

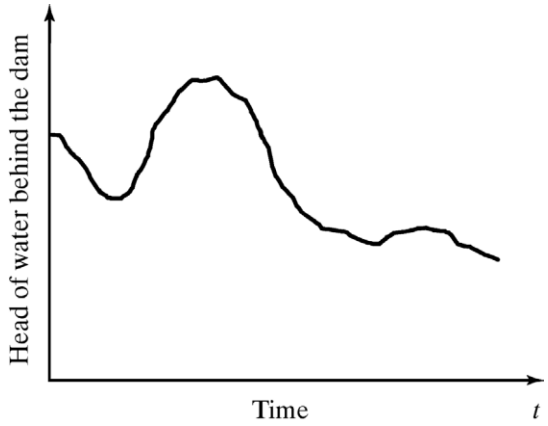
Discrete and Continues Systems

- A discrete system is one in which the state variables change only at a discrete set of points in time : Bank example



Discrete and Continues Systems (cont.)

A continues system is one in which the state variables change continuously over time: Head of water behind the dam



Model of a System

- To study the system
  - it is sometimes possible to experiments with system
    - This is not always possible (bank, factory,…)
    - A new system may not yet exist
- Model: construct a conceptual framework that describes a system
  - It is necessary to consider those aspects of the systems that affect the problem under investigation
    - simplification of the system (unnecessary details must be removed)
    - on the other hand, the model should be sufficiently detailed to permit valid conclusions to be drawn about the real system.

Model of a System

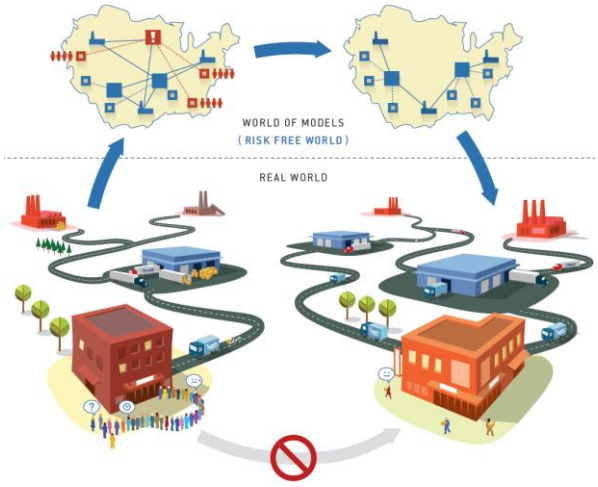
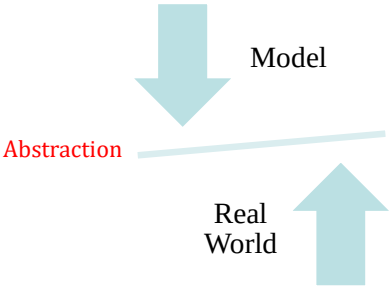
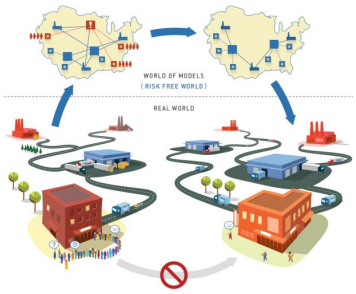


Figure source: "The Big Book of Simulation Modeling, Multimethod Modeling with AnyLogic 8", Authors: Andrei Borshchev, Ilya Grigoryev

Model of a System

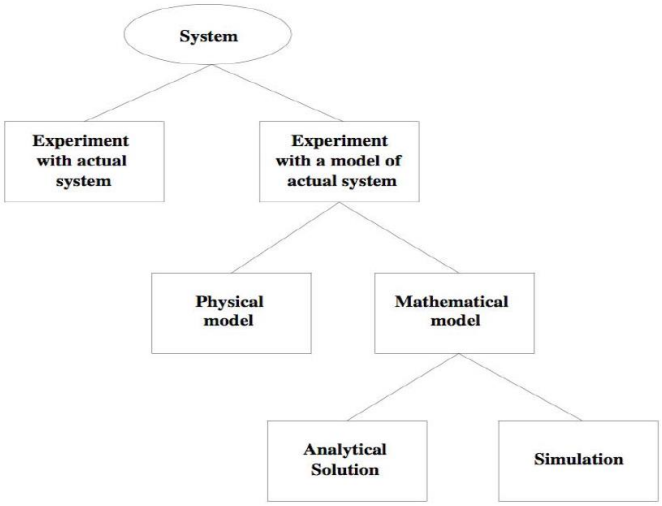




Abstraction Level?
Programming Language – Tool?

It's still more an art than a science.

Types of Models



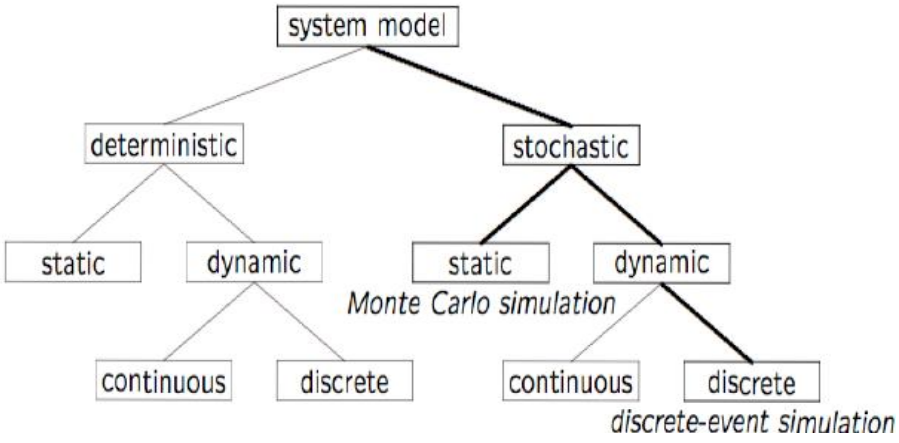
Characterizing a Simulation Model

- Deterministic or Stochastic
  - Does the model contain stochastic components?
  - Randomness is easy to add to a DES
- Static or Dynamic
  - Is time a significant variable?
- Continuous or Discrete
  - Does the system state evolve continuously or only at discrete points in time?
  - Continuous: classical mechanics
  - Discrete: queuing, inventory, machine shop models

Discrete-Event Simulation Model

- Stochastic: some state variables are random
- *Dynamic*: time evolution is important
- *Discrete-Event*: significant changes occur at discrete time instances

Model Taxonomy



DES Model Development

- How to develop a model:
  1) Problem formulation: by the policymakers or by the analyst
  2) Determine the goals and objectives
  3) Build a ***conceptual*** model
  4) Collect data: amount, type, accuracy, variety → complexity
  5) Convert the ***specification*** model into a ***computational*** model
  6) Verify: Did we build the model right?
  7) Validate: Did we build the right model?
  8) Experimental design: Alternatives? Runs? Periods?
  9) Production Runs and Analysis: estimate measures of performance
  10) More runs?
  11) Documentation and reporting
- Typically an iterative process

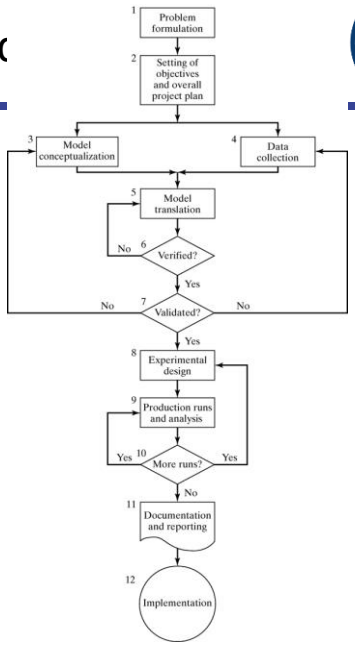
Three Model Levels

- Conceptual
  - Very high level
  - How comprehensive should the model be?
  - What are the *state variables,* which are dynamic, and which are important?
- Specification
  - On paper
  - May involve equations, pseudocode, etc.
  - How will the model receive input?
- Computational
  - A computer program
  - General-purpose PL or simulation language?

Verification vs. Validation

- *Verification*
  - Computational model should be consistent with specification model
  - Did we build the model right?
- *Validation*
  - Computational model should be consistent with the system being analyzed
  - Did we build the right model?
  - Can an expert distinguish simulation output from system output?
- Interactive graphics can prove valuable

Steps in Simulation Stud



Steps in Simulation Study

